

Milestone 01: Project Description and Objectives

CP317 - A - Group 6

Laurier Club Room Booking System

Team Members:

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Abstract

This project proposes the development of a web-based application to streamline the process of booking rooms on Laurier's campus for student clubs. Inspired by reservation systems used in the airline and car rental industries, the application will allow users to browse available rooms, check capacity and confirm reservations. Clubs will be able to view booking history, manage cancellations, and receive email notifications about their reservations. The project aims to enhance efficiency, reduce scheduling conflicts, and provide a centralized platform for room reservations, ultimately improving campus event organization.

Project Description

Currently, Laurier Student Unions clubs face challenges when booking rooms for meetings and events, often relying on manual processes or disorganized systems. Our solution is a room booking web-app that integrates availability, capacity and reservation management into a single, user-friendly platform. By borrowing functionality from existing reservation systems, the application will make room booking faster, more transparent and easier to manage.

Objectives

- Allow Laurier clubs to search and book rooms across campus
- Display important details such as room capacity and availability
- Provide a booking history feature for clubs to track past and upcoming reservations
- Send automated email notifications to users upon booking, cancellation or updates
- Reduce scheduling conflicts and improve transparency in the room booking process

Initial Product Backlog

First, for the user interface (UI) features:

- UI-1 (Search Availability): As a club member, I want to search rooms by date, time, capacity, and equipment so that I can find a suitable space for my event.
- UI-2 (Request Booking): As a club executive, I want to submit a room booking request so that I can secure a space for my club's meeting.
- UI-3 (Cancel Booking): As a club executive, I want to cancel an existing booking so that the room becomes available for others.

- UI-4 (View My Bookings): As a club member, I want to see a list of my upcoming and past bookings so that I can stay organized.
- UI-5 (Recurring Booking): As a club executive, I want to book a room for a recurring series of meetings so that I don't need to re-enter requests every week.

Next, for the administrator (ADM) features:

- ADM-1 (Approve/Decline Request): As an administrator, I want to approve or decline booking requests so that I can ensure fair and proper use of university spaces.
- ADM-2 (Set Booking Policies): As an administrator, I want to set limits on how many bookings a club can make per week so that no single group monopolizes the rooms.
- ADM-3 (Override Booking): As an administrator, I want to override an existing booking so that I can allocate a room for a priority university event.

For reporting and notifications:

- REP-1 (Room Utilization Report): As an administrator, I want to see reports on room usage by day and time so that I can identify high-demand and underused spaces.
- NOTIF-1 (Status Notifications): As a user, I want to receive email or in-app notifications when my booking is approved, declined, or modified so that I stay informed.

Ethical Implications

Ethical Issues and How They Are Addressed:

1. Data Privacy of Students:

We solve this in the design by storing only the name, student number, role, email and booking history. This is done in order to ensure we do not take any other private information that they have.

2. Fairness and Avoiding the Hoarding of Room Booking:

We solve this in the design by showing transparent rules about our room booking policy on the website, add quotas for advance bookings and rate limit so bad actors won't affect the booking for all students.

3. Accessibility for All Students:

We solve this by keeping the system simple and easy to use for everyone including students with disabilities. Clear layout, readable text and easy navigation to make the system inclusive.

4. Security of the System:

We solve this by making sure only authorized people can log in and view booking details. This prevents unauthorized users from accessing private information.

5. User Visibility and Control

An ethical challenge arises if users cannot see or manage how their data and bookings are handled. To address this, the system provides clear visibility of booking rules and personal booking history, giving all users control over their information. This respects their autonomy, supports informed decision-making, and builds trust in the system.

6. Compliance with Regulations

The system's design takes into account applicable Canadian privacy and accessibility laws, ensuring it meets both legal and ethical standards for handling user data.